

Satronix 

VISIT US AT **STALL H8-18** **ELECAMA-2012**
THE INDIA POWER FACTOR

Microcontroller Based
SOFT STARTERS
for Three Phase Induction Motors
POWER RANGE: 1.5 - 800 KW

SCR PROPORTIONAL CONTROL SYSTEMS
For Heating Applications
Input : Linear Control 4-20 mA or 0-10 VDC
Average O/P power 0 - 100 %

SCR ANALOG CONTROLLER
1 - Phase
Load: 25 - 330 VAC rms
45 - 480 VAC rms/ 10 - 30 Amps

DIGITAL SCR PROPORTIONAL CONTROLLER
2 & 3 Phase
Load: upto 450 VAC/90-205 Amps per phase

CAPACITOR SWITCHING MODULE
For Power Factor Correction
5 to 15 KVAR
LOAD : 25 - 600 VAC rms
25 to 90 Amps & Upto 265 Amps

SHORT CIRCUIT PROTECTED SSR
LOAD: 25 - 480 VAC rms
10 - 40 Amps

INTERFACE MODULES
Available with 4, 8, 16, 24, 32, 40 or 48 positions
Different configurations of DC - DC / DC - AC available

PCB MOUNTING SSR
DC - AC SSR (in built heat sink)
LOAD: 25 - 330 VAC / 3, 5, 7 Amps
DC - DC SSR
LOAD: 5 - 200 VDC / 3 Amps

TRANSFORMER SOFT START RELAY
(µP-based THYRISTORISED)
3-Phase & 1-Phase
LOAD 220V / 190VAC - 280VAC
440V / 320VAC - 480VAC
Current 10A - 90A

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purpose products targeted at specific applications.

2. There is a hairline gap between the RF and digital domains, and with the increasing demand for configurability and flexibility, this gap is going to further reduce over time.

3. The industry is also moving towards building automated test solutions (ATS) catering to specific test needs. These comprise a mix of RF, digital and general-purpose test equipment along with custom-built software. Cost and performance of these ATS could further be optimised by using both box and modular test platforms.

4. Work is underway to introduce new capabilities in high-performance and modular platforms so that both can complement each other to optimise the test platforms.

5. Another major advancement relates to closer integration of design and test. Samanta explains, "The test industry has not innovated as quickly, and many companies have chosen to invest more in their design tools than in test engineering tools. The consequence is, test engineers are typically outmatched when testing the latest software-centric DUT. *Pundits* in every major industry have envisioned solutions to bridge this gap. A closer look at the existing reconfigurable instrumentation architecture, which is the standard for many industries today, reveals a few common themes: a system-level approach, integration of design and test concepts, and extension of software architectures into FPGAs."

Tarun adds, "The next phase in integrating design and test is the ability of engineers to deploy design building blocks, known as intellectual property (IP) cores, to both the DUT and the reconfigurable instrument. This capability is called 'IP to the pin' because it drives user-defined software IP as close to the I/O pins of next-generation reconfigurable instruments as possible. The software IP includes functions/algorithms such as control logic, data acquisition, generation, digital protocols, encryption, math, RF and signal

processing.

"For example, MIMO SoC includes receivers, transmitters, converters, filters, switches and a processor. In addition, it features software IP such as coding, modulation, encryption and communication protocols. To fully validate the functionality of the highly integrated hardware and software sub-components of the SoC, engineers need system-level test capabilities to effectively emulate another communication device in the system. Because many of the IP blocks of the DUT and the test system are common, this presents an ideal case for concurrent design and test with IP reuse.

"The ability of a test engineer to directly embed the SoC design IP in the test instrumentation to perform system-level test can dramatically shorten design verification/validation and improve production test time and fault coverage. There are two key trends that will enable future reconfigurable test systems to deliver this IP-to-the-pin capability: the market shift toward FPGAs and availability of high-level software to program these."

6. There are also emerging multi-vendor IP ecosystems that feature IP cores from all major FPGA vendors as well as their software and instrumentation partners. The National Instruments' FPGA IPNet and the Cadence/Xilinx IP microsites are examples of these ecosystems. These contain hundreds of IP blocks and functions including the Xilinx CORE Generator, serial communication protocol cores, advanced encryption standard components as well as peer-to-peer streaming algorithms.

The way forward

RF design challenges and requirements posed by specific verticals are pushing the innovations further. Going forward, we can expect the emergence of more low-cost, sophisticated, customisable, multi-functional and precise equipment to meet the industry requirements. ■

The author is an executive editor at EYF